FIT PM12
Hardware test dossier

PM12-K7-3A17A1B402212223

Engineering diagnosis

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Sampling case resolved

Controlled phase sweeps and identical FPGA-only SRAM reloads. Functional capture qualification; complete board acceptance remains separate.

Prepared by _____

Reviewed by _____

Disposition / deviations _____

Signature and date _____

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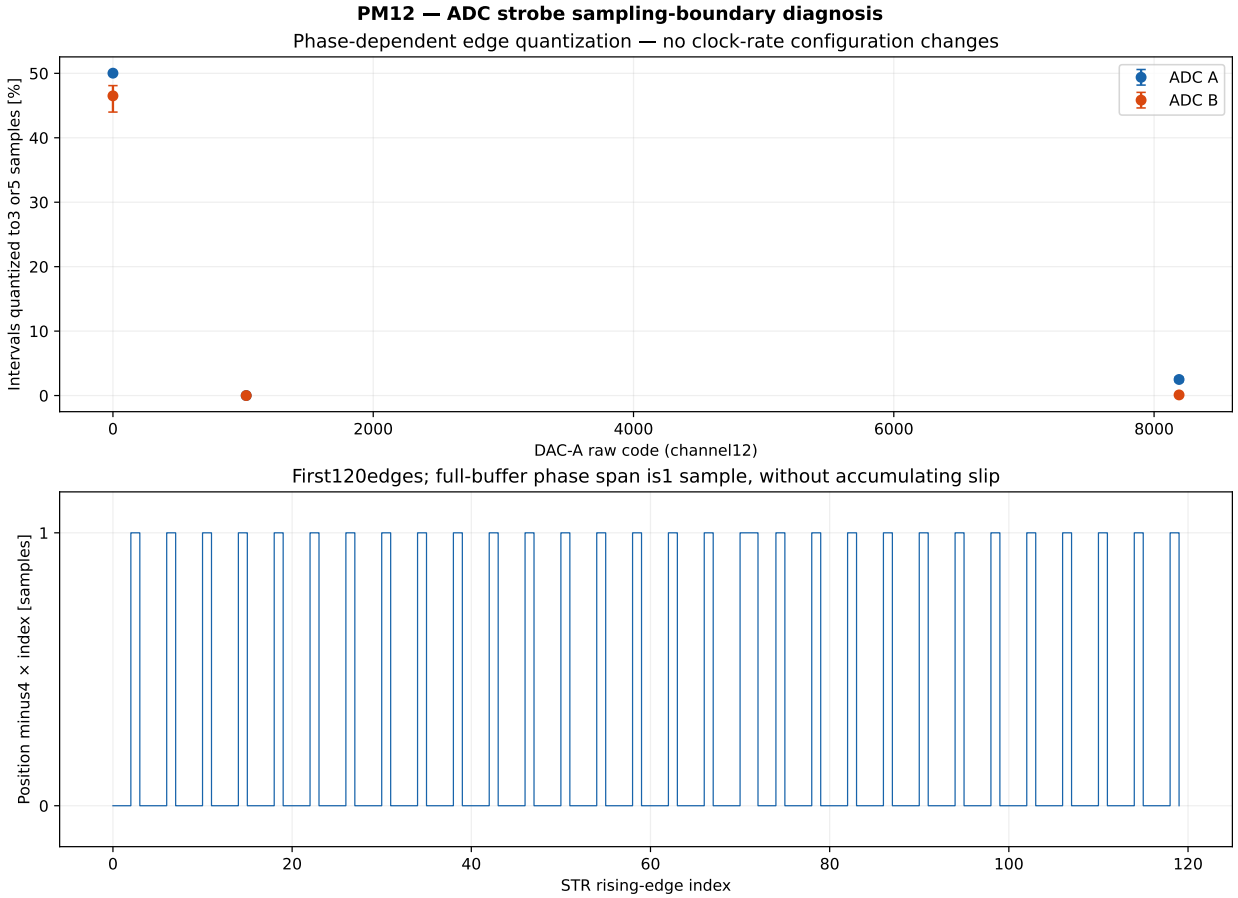
1 Finding and cause

The all4-tick STR interval requirement was invalid for asynchronous4x oversampling at a sampling boundary. STR and data use separate two-stage synchronizers. A small edge-position uncertainty produces intervals3/5 around the nominal4, without losing clock cycles. The clock trace alone does not measure physical jitter in picoseconds.

The same FPGA image was reloaded twice, with no CPLD EXTEST, CPLD programming, MCU reset, flash write or PLL profile change. Different channels then fell at the sampling boundary. The ADC word used for each measurement must remain equal across two consecutive raw samples.

2 Controlled phase repeat

Property	Observation
Unique retained raw buffers	38
ALIGNED	27
EDGE QUANTIZED	11
Maximum observed edge phase span	1 capture sample
Selected complete ADC words	All stable across two consecutive raw samples



DAC-A phase controls, ADC A/B; error bars show observed range across captures. No interpolation between measured settings. The lower panel displays the first120edges; the entire buffer has phase span1 sample.

3 FPGA-only reload control

Context	Buffers	Aligned	Quantized	Channels
before	72	66	6	12
reload0	72	71	1	10
reload1	72	66	6	3

Both banks and all12 digital ADC paths were captured three times per context. Every raw capture passed edge counts, bounded rising/falling phase, pulse-width and selected-word stability checks. CPLDs were not driven into EXTEST in this control.

4 Revised validity model

Check	Required condition
Capture framing	4096 words; correct EV tag; reserved bits zero
Edge count	1023 or1024 rising and falling edges
Edge spacing	Only3/4/5; cumulative phase spans at most1 sample on both polarities
Sampled pulse width	1..3 capture ticks
Data coherency	Selected pre-STR word equals its preceding full ADC-data sample
Strict alignment mode	Retained as a separate diagnostic; original strict FAIL evidence not changed

5 Full resident sequence repeated

Stage	Result
Hardware identity and PGOOD	PASS
CPU peripherals, SPI walking patterns and power telemetry	PASS
HDMI, all digital ADC paths and stimulated TDC	PASS
Gate delay versus internal DAC-A	PASS
ADC differential transfer and adjacent data-bit coverage	PASS

ADC raw validation	Result
pass	True
captures	514
raw_words	2105344
histograms_exact	True
ev_tags_exact	True
all_strobe_periods	None
quantized_buffers	369
selected_words_stable	True
max_allowed_edge_phase_span_samples	1
generations_contiguous	True

ADC	Visited codes	Non-rail samples	Bit/pair result
A	3099	241831	12bits /11pairs PASS
B	3053	241763	12bits /11pairs PASS

The257-point ADC A/B scan includes369quantized and145aligned captures. All514 histograms and EV tags match the raw buffers; generations are contiguous. After the two FPGA-only reloads, the final HDMI/ADC/TDC system test also passed without an unexpected SYS reset.

6 Limits and retained state

This diagnoses the sampling/validation case. It is not electrical jitter, clock spectral purity, source-synchronous timing-margin or complete ADC INL/DNL certification. Missing analog channels remain unmeasured. The same integrated image remains in SRAM; DAC-A raw0000, external DAC pair1V/1V, monitor inhibited, DIO outputs released, ADC bankA valid and TDC quiet/frozen. No configuration-flash or MCU/CPLD image writes.